

DOCUMENT 00

HYBRID SAFE CAPSULE

SCALABLE HYBRID LIFT PLATFORM

Master Engineering Specification — Final Expanded Baseline
Final Version 2.0 | Large Configuration (70 m Class)
Three-Layer Separable Architecture • Six-Layer Safety • Continuous CFRP Roll-Cage

DOCUMENT NUMBER	HSC-00-200-FINAL
DOCUMENT TYPE	Master Engineering Specification – Final Expanded Baseline
PROJECT	Hybrid Safe Capsule (Ultra-Safe Central Capsule Family)
AUTHORITATIVE BASELINE	Document 00 Hybrid Safe Capsule
BASELINE VERSION	Final Version 2.0
BASELINE DATE	September 2026
ENGINEERING MATURITY	Concept / Technical Baseline
DOCUMENT STATUS	COMPLETE BASELINE FOR DESIGN, SIMULATION, TESTING, PATENT EXTRACTION & INVESTOR REVIEW
CLASSIFICATION	CONFIDENTIAL — INTERNAL DESIGN USE ONLY
PROJECT FOUNDER	Amirbahador Atai
PREPARED BY	[AV]
DATE	September 2026

FINAL QUALITY GATE (MANDATORY)

CONDITIONALLY SUITABLE FOR CONTROLLED ENGINEERING REVIEW AND INVESTOR DISCUSSION ONLY —
NOT VERIFIED, NOT VALIDATED, NOT QUALIFIED, NOT CERTIFIED, AND NOT APPROVED FOR RELEASE [AV]

1. PURPOSE AND SCOPE

This Document 00 establishes the authoritative technical baseline for the Hybrid Safe Capsule (HSC), a scalable hybrid-lift airborne safety platform. It defines geometry, mass properties, buoyancy architecture, distributed propulsion, the continuous carbon-fiber roll-cage of the upper occupant capsule, the six-layer safety architecture, and the three-layer separable system (Lower Propulsion Layer – Central Helium Layer – Upper Occupant Capsule). The document is written to support controlled engineering review, preliminary analysis, patent extraction, investor evaluation, and the subsequent generation of the full Master Document Tree.

All numerical values presented herein are design targets or concept-level estimates derived from first-order aerospace engineering relationships. They are not verified test results, validated performance data, or certification evidence. Where data are incomplete or require further analysis they are explicitly marked [AV] (assumption requiring verification), [ND] (not yet determined), or [OPEN] (open design space). This discipline is mandatory for all downstream documents that reference this baseline. Any silent upgrade of a design target into a verified result is prohibited.

2. STRATEGIC VALUE PROPOSITION FOR INVESTORS

The Hybrid Safe Capsule addresses a structural and persistent gap in the aerospace and emergency-response markets: the absence of a large-scale, runway-independent, highly survivable aerial platform that combines passive buoyant lift with precise powered control and layered crew protection. Conventional helicopters and emerging eVTOL aircraft carry high single-point failure risk in the event of power loss or structural compromise. Pure airships and hybrid airships provide buoyant lift but typically lack the rapid vertical authority, precision hover, and progressive failure isolation required for high-value crew and cargo missions.

The HSC architecture is deliberately engineered to occupy the intersection of these domains. Helium supplies a large passive lift reserve that reduces continuous power demand and creates a natural graceful-degradation path. Distributed ducted fans supply the residual lift, attitude control and landing authority that pure buoyancy cannot provide. The upper occupant capsule is structurally independent, protected by a continuous multi-axis carbon-fiber roll-cage, and equipped with its own recovery system so that crew survival does not depend on the integrity of the entire vehicle.

Key attributes of investor relevance:

- Passive lift reserve materially reduces continuous power demand and improves emergency endurance relative to pure powered-lift vehicles of similar mass.
- Three-layer physical separation provides progressive failure isolation and multiple crew-survival paths that conventional aircraft do not offer.
- Internal continuous carbon-fiber roll-cage in the occupant capsule delivers multi-axis crash, rollover and inversion protection as a primary structural element, not an after-thought.
- Scalable geometry family (approximately 30 m to 100 m class) enables a coherent product line from tactical to heavy industrial and dual-use roles.
- Large and defensible patent surface across buoyancy–structure integration, three-layer separation sequencing, roll-cage architecture, multi-mechanism release systems and hybrid control logic.
- Clear dual-use and civil market pathways: industrial heavy-lift logistics, disaster response and mass evacuation, offshore personnel transport, temporary elevated infrastructure, and defense logistics under high-survivability requirements.

Because the baseline is written in conservative, traceable language and explicitly marks every unverified number, the document itself is suitable as a controlled engineering foundation for investor due diligence, patent-family extraction and the subsequent generation of a full 500-document engineering package.

3. SYSTEM OVERVIEW AND DESIGN PHILOSOPHY

The Hybrid Safe Capsule is a large-scale airborne safety platform that combines helium buoyancy with distributed powered lift. The architecture prioritizes survivability, controlled vertical operations, graceful degradation under partial failure, and an independent upper occupant capsule capable of separation and autonomous recovery. Unlike conventional aircraft, the concept emphasizes stationary endurance, controlled vertical movement, passive safety

margins and load-bearing stability over high transit speed.

Core design principles locked into this baseline:

- Hybrid lift: helium supplies the majority of static lift; fans supply control, trim, residual lift and emergency authority.
- Distributed propulsion and multi-layer redundancy to eliminate single-point catastrophic failure modes.
- Three-layer separable architecture (Propulsion Deck / Helium Module / Occupant Capsule) with independent recovery capability for the crew cell.
- Internal continuous carbon-fiber roll-cage protecting the crew volume against multi-axis impact, rollover and inversion.
- Scalable geometry family sharing a common engineering language, separation philosophy and safety architecture.
- Safety-first philosophy: controlled descent and occupant protection take precedence over pure aerodynamic efficiency or maximum speed.

4. THREE-LAYER SEPARABLE ARCHITECTURE

The platform is organized into three physically distinct and functionally independent layers that can be separated in a controlled, progressive sequence under emergency conditions. This three-layer architecture is the primary structural and safety contribution of the Hybrid Safe Capsule concept and is intended to form a core patent family.

4.1 Layer Definitions

Layer	Designation	Primary Function	Key Features
Layer A	Lower Propulsion / Rotor Deck	Powered residual lift, primary fans, landing-gear interface	4 × 20 m primary fans + 16 distributed landing legs
Layer B	Central Helium Buoyancy Module	Static buoyant lift, multi-cell helium volume, primary structural ring	~110 000 m ³ segmented helium, CFRP outer ring & radial beams
Layer C	Upper Occupant Safety Capsule	Crew volume, flight controls, roll-cage protection, independent recovery	Continuous CFRP roll-cage, 16 dedicated parachutes, life support

4.2 Separation Philosophy

Separation is progressive and hierarchical. The design intent is that severe damage or loss of control authority in one layer does not automatically destroy the survival capability of the occupant capsule. Each separation interface is equipped with multi-mechanism release hardware so that no single electrical, software or single-point mechanical failure can prevent release when a valid separation command is generated. The upper capsule (Layer C) is designed to remain a fully functional survival cell after separation, with its own parachute array, independent power, attitude control and life-support provisions.

4.3 Nominal Emergency Separation Sequence

- Stage 1 – Propulsion Deck Isolation (Layer A): If fire, severe vibration, uncontained engine/fan failure or structural compromise is detected in the lower propulsion deck, Layer A may be jettisoned or isolated. The helium module (Layer B) and occupant capsule (Layer C) remain coupled. This removes a major mass and hazard source while preserving buoyant lift and the crew cell.
- Stage 2 – Helium Module Release (Layer B): Under sustained loss of control authority, progressive structural failure of the main platform, or when residual mass must be reduced to improve recovery dynamics, the central helium module can be released. The upper occupant capsule continues on its own recovery path. Release of Layer B is a mass-reduction and simplification step, not a primary survival step for the crew.
- Stage 3 – Occupant Capsule Autonomous Recovery (Layer C): The upper safety capsule separates cleanly from the remaining structure, deploys its dedicated 16-parachute array, activates independent emergency power and life-support, and performs controlled descent under its own recovery system. The continuous carbon-fiber roll-cage remains the primary structural protection for the crew throughout this phase and after landing.

The architecture is designed so that the occupant capsule can complete a successful recovery even if Stages 1 and 2 are only partially executed or are omitted. Full three-stage separation is the highest-survivability path; partial sequences still provide progressive protection and are therefore preferred to a single all-or-nothing separation event.

5. GEOMETRIC CONFIGURATION (LARGE BASELINE)

The reference large configuration is a square-planform platform. Primary dimensions and related geometric parameters are locked for this baseline as follows. All subsequent CAD and analysis work shall respect these envelope values unless a formal baseline change is approved.

Parameter	Value	Notes
Platform Width / Length	70 m × 70 m	Square planform reference
Overall Height	32 m	Structural height + landing gear envelope
Helium Chamber Volume (approx.)	~110 000 m ³	Segmented multi-cell arrangement
Upper Capsule Envelope (concept)	~18 m × 12 m class	Independent structural module [AV]
Usable Volume Efficiency Target	~85 %	Geometry vs structure/duct/payload trade [AV]

The geometry is chosen to maximize effective buoyancy volume while preserving structural corridors for radial beams, fan ducts, payload decks and the central/upper capsule bay. Exact internal volume allocation, roll-cage clearances, seat packaging and egress path geometry remain subject to detailed CAD definition and human-factors validation [OPEN]. Early packaging of the continuous roll-cage is a critical-path activity because it directly affects usable crew volume and egress routes.

6. MASS AND LIFT BALANCE

First-order mass and buoyancy estimates for the 70 m class baseline are summarized below. Values are concept-level targets and include engineering margins for envelope mass, thermal effects and structural supports. They must not be treated as verified empty-weight or performance figures.

Item	Estimate	Remark
Structural mass (CFRP-dominant)	~120 t	~80 % composite by mass target
Helium net lift capacity	~112 t	Based on ~110 000 m ³ usable volume
Effective residual mass on fans	~8 t	Fans primarily control & residual lift
Net buoyancy (sea-level reference)	~1.047 kg/m ³	Air 1.225 – He 0.178 kg/m ³
Practical usable helium lift	~112 t	After realistic system losses [AV]

6.1 Preliminary Mass Allocation by Layer (Concept Target)

The following first-order allocation is provided for guidance only. It is not a verified mass budget and must be refined through detailed CAD and subsystem definition [AV]. The allocation is intentionally conservative and includes growth margin.

Layer	Concept Mass Target	Primary Contributors
Layer A – Propulsion Deck	~45–55 t	Primary fans, motors, landing legs, structure, power distribution
Layer B – Helium Module	~50–60 t	CFRP outer ring, radial beams, helium cells, systems, inter-layer interfaces
Layer C – Occupant Capsule	~15–25 t	Continuous roll-cage, shell, seats, life support, recovery systems, independent power
Total structural target	~120 t	Consistent with overall baseline; growth reserve included

Because helium carries the majority of structural weight, the propulsion system is sized primarily for stability, wind compensation, controlled vertical motion and emergency residual lift rather than full vehicle weight support. This architecture reduces continuous power demand and improves emergency endurance margins. Detailed

mass-properties models with subsystem allocation and center-of-gravity tracking for each of the three layers are required as a next-step activity [OPEN].

7. BUOYANCY SYSTEM ARCHITECTURE

The buoyancy system consists of segmented helium chambers integrated into the central structural volume (Layer B). Compartmentalization is a mandatory safety feature: it limits the consequence of localized puncture, leakage or thermal damage. Pressure regulation, thermal expansion management and continuous leak detection are required elements of the detailed design and will be specified in downstream buoyancy-system documents.

Key buoyancy design features locked into this baseline:

- Multi-cell helium arrangement with independent containment zones and damage isolation so that loss of one or several cells does not produce immediate catastrophic loss of lift.
- Target usable volume efficiency of approximately 85 % of the geometric envelope, balancing lift volume against structure, ducts and payload [AV].
- Structural integration of the chambers with carbon-fiber load rings and radial beams so that the buoyancy volume participates in global load paths rather than acting as a passive bag.
- Ballast and trim management coordinated with active fan control to maintain attitude stability under asymmetric loading or cell loss.
- Thermal expansion and differential-pressure management across the full operating altitude and temperature envelope [OPEN].
- Graceful degradation philosophy: the platform must remain controllable or passively recoverable after credible partial loss of buoyancy.

8. PROPULSION ARCHITECTURE

The propulsion system is distributed and dual-role. Primary vertical lift fans located in Layer A provide residual lift, hover support and controlled descent authority. Outer-ring stabilization and control fans provide attitude control, yaw authority, lateral trim and wind rejection. The distributed arrangement is intentional: failure of any single fan unit must not result in immediate loss of controllability of the remaining vehicle.

Subsystem	Qty	Diameter	Primary Function
Primary Lift Fans (Layer A)	4	20 m class	Vertical residual lift, hover, controlled descent
Stabilization / Control Fans	9	9 m class	Roll, pitch, yaw, lateral trim, wind rejection

Thrust allocation algorithms, control laws, motor thermal management and single-engine-out (or single-fan-out) performance remain to be developed in subsequent propulsion and flight-control documents [OPEN]. The baseline requirement is that the vehicle must retain controllable vertical and attitude authority after loss of any one primary fan and any one stabilization fan under design environmental conditions.

9. INTERNAL CARBON-FIBER ROLL-CAGE (UPPER OCCUPANT CAPSULE)

The upper occupant safety capsule (Layer C) incorporates a continuous multi-axis carbon-fiber roll-cage as its primary crashworthiness, rollover and inversion-protection structure. The roll-cage is not an add-on cage bolted into an existing shell; it is an integral primary structure that forms the protected crew cell and participates directly in the load paths of the capsule. This is one of the highest-value differentiators of the Hybrid Safe Capsule relative to conventional crew compartments.

9.1 Design Intent and Functional Requirements

- Provide continuous structural continuity around the entire crew volume in all six primary axes (longitudinal, lateral, vertical, roll, pitch and yaw).
- Maintain a protected occupant cell under multi-axis impact, rollover, inversion and water-landing loads so that the crew volume does not collapse or invert uncontrollably.
- Transfer separation loads, parachute opening loads and landing loads into the outer capsule shell and floor without local collapse of the crew cell.

- Serve as the primary attachment backbone for energy-absorbing seats, restraint systems, internal equipment and emergency systems.
- Remain fully functional as a protective structure after complete separation of Layer C from Layers A and B.
- Support side-door and roof-hatch egress paths without creating structural discontinuities that would compromise the continuous load path.

9.2 Structural Concept and Geometry

The roll-cage is conceived as a continuous carbon-fiber tubular and beam network with titanium or high-strength metallic nodes at primary intersections and attachment points. Primary tube diameter is targeted in the 100–140 mm class, with local reinforcement (additional plies, metallic doublers or larger cross-sections) at high-load interfaces such as parachute attachment points, seat rails and separation fittings [AV]. The cage is bonded and mechanically fastened to a continuous structural floor and to the upper structural ring of the capsule so that floor, cage and outer shell act as a single load-bearing cell.

No floor hatch is provided in the baseline configuration. Egress is via two side doors and one roof hatch. This decision preserves a continuous floor load path, improves under-body impact resistance and simplifies the integration of energy-absorbing seat structures. Detailed three-dimensional clearance studies after installation of the roll-cage tubes, seats, control consoles and life-support equipment are required to confirm usable internal volume, head clearance and unobstructed egress paths [OPEN]. Early CAD packaging of the roll-cage is therefore a critical-path activity and must be completed before any detailed seat or console design is frozen.

9.3 Materials, Safety Factor and Verification

A structural safety factor target of ≥ 2.8 on ultimate loads for the primary roll-cage members is adopted as a concept-level requirement [AV]. Primary material is high-modulus aerospace carbon-fiber composite with a toughened epoxy or equivalent resin system selected for damage tolerance and environmental resistance. Metallic nodes (titanium preferred for strength-to-weight and corrosion resistance) are used at concentrated load points to manage bearing, peel and fastener loads. Full finite-element analysis of the integrated cage + floor + shell under crash, separation, parachute-opening and inverted-landing load cases is a mandatory next-step activity before any detailed design freeze.

10. MULTI-LAYER SAFETY ARCHITECTURE

Safety is treated as an embedded multi-layer architecture rather than an accessory system. The Hybrid Safe Capsule implements six independent safety layers that operate both in parallel (for continuous protection) and in sequence (for progressive recovery). No single layer is relied upon as the sole means of occupant survival. This six-layer model is intended to be a core element of the safety case and of the patent portfolio.

10.1 Six Safety Layers – Baseline Definition

L	Name	Primary Function
1	Sensing & Command	Independent sensing, continuous health monitoring, automatic and crew-commanded separation and recovery triggers
2	Pyrotechnic / Active Release	Pyro-shear bolts, cutters and active release devices for each separation interface with dual-redundant initiation
3	Pure Mechanical Override	Dual-operator mechanical release independent of electrical power, software and pyrotechnics
4	Cold-Gas / Auxiliary Actuation	Cold-gas or stored-energy actuators for backup door, hatch and selected release and deployment functions
5	Impact Absorption & Structure	Continuous CFRP roll-cage, energy-absorbing seats, progressive landing legs, global structural load paths
6	Post-Landing Survival	Parachute arrays, flotation / self-righting, 72-hour life-support target, dual-band location beacons

10.2 Independence and Common-Cause Rules

- Layer 3 (pure mechanical override) must function with total loss of electrical power, software and pyrotechnic initiation capability.
- Layer 5 (structure and energy absorption) must protect occupants even if parachute deployment is delayed, partial or asymmetric.
- Layer 6 (post-landing survival) must provide a breathable atmosphere, basic thermal control and location capability for a minimum of 72 hours after landing or water entry [AV].
- No single sensor, processor, power bus, data bus or pyrotechnic channel may be a common-cause failure for more than one critical safety function across different layers.
- Separation and recovery sequences shall be designed so that automatic paths and crew-commanded paths are both available and are not mutually exclusive.

11. REPRESENTATIVE FAILURE SCENARIOS AND RECOVERY PATHS

The following scenarios illustrate how the three-layer architecture and six-layer safety system are intended to provide progressive protection. They are conceptual and do not constitute verified test cases or certification evidence [AV]. The purpose is to make the recovery logic explicit for engineering and investor review.

11.1 Scenario A – Propulsion Deck Fire / Uncontained Fan Failure

Detection by Layer 1 sensors triggers automatic or crew-commanded isolation/jettison of Layer A (Stage 1). Layers B and C remain coupled. Helium lift is preserved. The occupant capsule continues under buoyant support with residual fan authority from any remaining control fans. If further degradation occurs, Stage 2 (helium module release) and Stage 3 (capsule separation) remain available. The continuous roll-cage (Layer 5) and post-landing systems (Layer 6) protect the crew after any subsequent separation. This scenario demonstrates the value of physical separation of the highest-hazard propulsion elements from the crew cell.

11.2 Scenario B – Progressive Structural Failure of Main Platform

Sustained loss of control authority or progressive structural compromise of the main platform triggers Stage 2 (helium module release) and/or Stage 3 (occupant capsule separation). Layer C deploys its dedicated 16-parachute array, activates independent power and life support, and descends under roll-cage protection. Partial sequences still leave the crew with a functional survival cell. The design explicitly prefers progressive isolation over a single all-or-nothing separation event.

11.3 Scenario C – Partial Parachute Deployment or Asymmetric Landing

Even with delayed, partial or asymmetric parachute performance, Layer 5 (continuous roll-cage + energy-absorbing seats + progressive landing legs) is required to keep the crew volume intact. Layer 6 provides post-landing survival (atmosphere, thermal control, location) for the 72-hour target. The design explicitly does not rely on perfect parachute performance as the sole survival path. This is a key differentiator relative to systems that treat parachutes as the primary and only recovery means.

11.4 Scenario D – Combined Power and Buoyancy Degradation

In the event of simultaneous partial loss of fan authority and partial loss of helium lift (for example multiple cell leakage plus single-fan-out), the remaining systems must still provide a controllable or passively recoverable path to the surface. The baseline requires that the combination of residual buoyancy, residual fan thrust and the six safety layers keep the occupant risk within the design envelope. Quantitative demonstration of this combined case is an open analysis item [OPEN].

12. SAFETY AND EMERGENCY HARDWARE SUMMARY

Safety Element	Configuration (Large Baseline)
Main Structure Parachutes	16 units, symmetrically distributed across the upper structure
Capsule Parachutes (Layer C)	16 units dedicated exclusively to the upper occupant capsule
Total Parachute Count	32

Landing Legs	16 hydraulic / hybrid energy-absorbing legs
Layer Separation System	Multi-mechanism: EMLG + pyro-shear + dual-operator mechanical override
Flight Control Redundancy	Triple-redundant pathways with fault isolation [AV]
Internal Roll-Cage	Continuous multi-axis CFRP cage with metallic nodes, SF ≥ 2.8 target
Life Support (post-separation)	Minimum 72-hour endurance target (O ₂ , CO ₂ removal, thermal) [AV]
Location & Recovery Aids	Dual-band beacons, high-visibility markings, flotation / self-righting [AV]

13. LANDING SYSTEM AND STRUCTURAL ARCHITECTURE

The landing system comprises 16 distributed hydraulic or hybrid energy-absorbing legs mounted to reinforced nodes on the outer structural ring (primarily associated with Layer A). Approximate leg envelope is 3 m width $\times$ 1 m height in the retracted state. The large footprint reduces ground pressure and improves stability on uneven or semi-prepared surfaces. Progressive energy absorption is required so that landing loads transmitted into the upper capsule and roll-cage remain within the structural design envelope. Detailed energy-absorption characteristics, terrain-adaptation performance and combined platform-plus-capsule landing loads remain to be defined through simulation and testing [OPEN].

Primary structural material is carbon-fiber composite, targeted at approximately 80 % of structural mass. Secondary components use aerospace aluminum alloys and titanium inserts at high-load interfaces. The global load-path architecture employs an outer ring, radial beams and concentric load-distribution rings. The upper capsule integrates the continuous carbon-fiber roll-cage as the primary crew-cell structure. Fail-safe philosophy: localized structural damage shall not produce immediate global collapse of the protected crew cell.

14. COMPARISON WITH CONVENTIONAL HELICOPTERS AND AIRSHIPS

The following high-level comparison is conceptual and intended to clarify the intended market position of the Hybrid Safe Capsule. It is not a verified performance claim [AV]. The purpose is to show why a new architectural class is proposed rather than an incremental improvement of existing vehicle types.

Attribute	Conventional Helicopter	Pure / Hybrid Airship	Hybrid Safe Capsule (HSC)
Primary lift source	Powered rotor	Helium buoyancy	Helium + distributed fans
Single-point failure risk	High (power / rotor)	Moderate (envelope)	Reduced by three-layer separation
Crew protection after major failure	Limited (autorotation / crashworthiness)	Limited (envelope integrity)	Dedicated separable capsule + roll-cage + 16 parachutes
Runway independence	Yes (VTOL)	Yes	Yes (VTOL + buoyant reserve)
Progressive failure isolation	Low	Low–moderate	High (three physical layers)
Primary design priority	Speed / payload / agility	Endurance / efficiency	Survivability + controlled vertical ops
Post-separation crew cell	Not applicable	Not standard	Independent Layer C with own recovery

The HSC is not intended to replace high-speed helicopters or long-endurance pure airships in their core roles. It is intended to occupy the high-survivability, runway-independent, controlled-vertical niche where crew and high-value cargo protection after major failure is the dominant requirement. This positioning is central to the investment and patent strategy.

15. PRELIMINARY PATENT-FAMILY SURFACE (CONCEPT)

The following list identifies high-level concept areas that are considered strong candidates for patent-family extraction. No claims have been filed; this is a planning surface only [AV]. A formal preliminary patent-family

assessment document should be prepared as a critical-path activity immediately after geometry and separation interface freeze.

- Three-layer physically separable hybrid-lift architecture with progressive emergency separation sequence and multi-mechanism release interfaces.
- Continuous multi-axis carbon-fiber roll-cage integrated as primary crew-cell structure in a separable aerial capsule, including continuous floor load path and side/roof egress only.
- Multi-mechanism release system (electromechanical + pyrotechnic + dual-operator mechanical override) specifically for aerial layer separation under degraded power and software conditions.
- Six-layer independent safety architecture coordinated with progressive physical layer separation, including explicit independence and common-cause rules.
- Hybrid buoyancy + distributed fan control with residual-mass sizing philosophy and single-fan-out controllability requirements.
- Post-separation autonomous recovery sequence for an occupant capsule carrying independent power, life support, attitude control and dedicated parachute array.
- Integration of continuous roll-cage with energy-absorbing seats and progressive landing loads so that structural protection remains effective even under partial parachute performance.

These seven concept areas are expected to generate multiple claim sets and supporting embodiments. The conservative language of this baseline is intended to support rather than prejudice later formal patent drafting.

16. ENERGY SYSTEM AND SCALABILITY

A hybrid energy architecture is the preferred concept baseline: a turbo-generator provides sustained electrical power for continuous operation while a battery subsystem supplies high instantaneous power during peak demand (hover, takeoff, emergency maneuvers). The upper occupant capsule carries an independent emergency power source sufficient to support recovery systems, attitude control, communications and life support after separation. Exact power levels, fuel capacity, electrical distribution architecture and thermal management remain reserved for subsequent power-system documents [OPEN]. Short-duration emergency hover or controlled-descent capability on battery support is a design target (order of several minutes) [AV].

The architecture is intentionally scalable across a diameter family from approximately 30 m to 100 m. Larger variants increase both helium lift and required fan capacity while preserving the same hybrid-lift principle, three-layer separation concept, continuous roll-cage philosophy and six-layer safety approach. This creates a coherent product family that can support progressive market entry, common training and maintenance practices, and a unified intellectual-property portfolio suitable for licensing or sale.

17. PRIMARY APPLICATION DOMAINS

The Hybrid Safe Capsule is intended to serve multiple high-value roles in which runway independence, survivability and controlled vertical operations are more important than high transit speed:

- Heavy-lift aerial logistics and industrial crane roles where conventional helicopters reach payload or endurance limits and fixed-wing aircraft require runways.
- Disaster response and mass evacuation from flood zones, wildfire regions, earthquake-damaged urban areas and conflict-affected zones where ground infrastructure is unavailable.
- Offshore and remote-site personnel transport with enhanced survivability relative to conventional helicopter operations.
- Temporary elevated command posts, medical support nodes, communications relays and logistics transfer points.
- Defense and dual-use logistics under high-survivability and runway-independent requirements.

18. ASSUMPTIONS, LIMITATIONS AND BOUNDARIES

This baseline is a concept-level technical specification. The following limitations apply and must be respected by all downstream documents and by any investor or partner communication that references this document:

- All mass, lift, power and endurance figures are first-order estimates; they require detailed CAD, FEA, CFD and subsequent test confirmation.

- No flight-test, structural-test, propulsion-test or human-factors validation data are claimed or implied by this document.
- Certification pathway, regulatory special conditions and formal safety-case approval are not yet established.
- Patent concepts identified in Section 15 or in related documents are not issued patents.
- Environmental operating envelope (wind, temperature, icing, water landing, gust response, bird strike, etc.) requires further definition.
- Values marked [AV] are assumptions requiring verification; [ND] indicates not yet determined; [OPEN] indicates open design space.
- This document does not constitute proof that the Hybrid Safe Capsule has been designed, built, tested, qualified, certified or proven safe.

19. KEY SPECIFICATION SUMMARY (TRACEABLE)

Attribute	Large Baseline (Final v2.0)
Platform size	70 m × 70 m × 32 m
Structural mass target	~120 t (CFRP dominant, ~80 % composite)
Helium volume / practical lift	~110 000 m ³ / ~112 t
Residual mass on fans	~8 t
Primary lift fans	4 × 20 m class (Layer A)
Stabilization fans	9 × 9 m class
Parachutes (total)	32 (16 main structure + 16 capsule)
Landing legs	16 energy-absorbing units
Architecture	Three-layer separable (A / B / C)
Occupant protection	Continuous multi-axis CFRP roll-cage + six safety layers
Safety philosophy	Six independent layers + progressive three-stage separation
Engineering maturity	Concept / Technical Baseline — not verified or certified

20. DOCUMENT CONTROL AND RECOMMENDED NEXT STEPS

This Document 00 (Final Version 2.0) is the locked technical baseline for the Hybrid Safe Capsule large configuration. Subsequent documents in the Master Document Tree shall derive numerical and architectural content from this baseline without silent modification or upgrade of targets into verified results. Any change to a locked value requires a formal baseline revision.

Recommended immediate next steps (critical path):

- Detailed parametric CAD model of the 70 m platform, the three layers and the upper-capsule continuous roll-cage (geometry and packaging lock).
- First-order FEA of primary load paths, roll-cage, separation interfaces, parachute attachment points and landing-leg hard points.
- Updated mass-properties model with full subsystem allocation and center-of-gravity analysis for each of the three layers.
- Critical-path safety documents: Six-Layer Safety Architecture, Layer Separation Architecture, Roll-Cage Structural Specification, Parachute System Specification.
- Preliminary patent-family extraction based on the surface listed in Section 15.
- Definition of the environmental operating envelope and preliminary gust / wind-load / water-landing cases.
- Investor-oriented summary package derived strictly from this locked baseline, suitable for controlled disclosure under NDA.

END OF DOCUMENT 00 — FINAL VERSION 2.0 (HYBRID SAFE CAPSULE)

*CONDITIONALLY SUITABLE FOR CONTROLLED ENGINEERING REVIEW AND INVESTOR DISCUSSION ONLY —
NOT VERIFIED, NOT VALIDATED, NOT QUALIFIED, NOT CERTIFIED, AND NOT APPROVED FOR RELEASE [AV]*

Founder: Amirbahador Atai | Hybrid Safe Capsule Project | September 2026 | CONFIDENTIAL